Simplification Plan for pmsimstats-simple

Contents

1 Overview **2**

2 Current Complexity in pmsimstats2025 **2**

3 Proposed Simplifications **2**

3.1 Data Generation . . . 2
3.1.1 Covariance Structure . . . 2
3.1.2 Response Components . . . 2
3.2 Statistical Analysis . . . 3
3.2.1 Model Choice . . . 3
3.2.2 Handling Repeated Measures . . . 3
3.3 Mathematical Specification of the Interaction Test . . . 4
3.3.1 Notation . . . 4
3.3.2 The Fundamental Distinction: Within-Subject Drug Variation . . . 4
3.3.3 Method 1: ANCOVA on Phase-Mean Change Scores . . . 5
3.3.4 Method 2: Paired t-test on Change Scores . . . 6
3.3.5 Method 3: GEE (Generalized Estimating Equations) . . . 7
3.3.6 Method 4: Random Intercept LMM (`lme4::lmer`) . . . 8
3.3.7 Method 5: Simple Linear Regression (Ignoring Clustering) . . . 9
3.3.8 Method 6: Repeated Measures ANOVA (Biomarker Dichotomized) . . . 10
3.3.9 The OL Design: A Fundamentally Different Interaction Test . . . 12
3.3.10 Summary: Interaction Test Precision Across Designs and Methods . . . 12
3.4 Trial Designs (Non-Simplifiable) . . . 13
3.5 Carryover Effects . . . 13
3.6 Censoring/Dropout . . . 14
3.7 Parameter Space . . . 14

4 Implementation Strategy **14**

5 Code Architecture **14**

6 Expected Outcomes **15**

7 Literature Support for Simplifications **15**

7.1 Key Reference: Hendrickson et al. (2020) . . . 15
7.2 Analysis Method Comparisons . . . 15
7.3 ANCOVA for RCTs . . . 16

7.4 Biomarker-Treatment Interaction Power 16
7.5 N-of-1 Power Analysis 16
7.6 Implications for Simplification 16

8 Decision Log **16**

9 References **16**

1 Overview

This document outlines possible simplifications to the N-of-1 trial simulation for biomarker-moderated treatment response. The goal is to reduce complexity while preserving the core scientific question: **What is the statistical power to detect a biomarker × treatment interaction across different trial designs?**

2 Current Complexity in pmsimstats2025

Component	Current Implementation	Complexity
Covariance structure	Full MVN with AR(1), cross-component correlations	High
Data generation	Two-stage conditional sampling from joint distribution	High
Analysis model	Mixed-effects (lme) with CAR(1) correlation	High
Carryover	Exponential decay via half-life parameter	Medium
Censoring	5 different dropout/censoring patterns	Medium
Response components	BR, ER, TR with separate random effects per timepoint	High

3 Proposed Simplifications

3.1 Data Generation

3.1.1 Covariance Structure

Current: Joint multivariate normal for BR, ER, TR random effects across all timepoints, plus biomarker and baseline, with AR(1) temporal structure and cross-component correlations.

Simplified options:

Option	Description	Pros	Cons
Independent errors	response = signal + ε	Trivial to implement	Ignores repeated measures
Compound symmetry	Equal correlation between all timepoints	Simple, one parameter	Unrealistic for longitudinal
Random intercept only	$Y = \text{signal} + u_i + \varepsilon_{ij}$	Standard mixed model	No temporal correlation
AR(1) residuals only	Single AR(1) process, no component separation	Moderate complexity	Loses BR/ER/TR decomposition

Recommendation: Start with **random intercept + independent errors**. This captures between-subject variability without temporal complexity.

```
u_i <- rnorm(1, 0, sigma_between)
epsilon <- rnorm(n_timepoints, 0, sigma_within)
response <- baseline + signal + u_i + epsilon
```

3.1.2 Response Components

Current: Separate BR, ER, TR components each with own random effects.

Option	Description	Implementation
Single treatment effect	Collapse BR into one effect	`effect = on_drug * beta * (1 + bm_mod * biomarker)`
No ER component	Remove expectancy response	Reduces parameters by 1/3
No TR component	Remove time/regression to mean	Assumes stable baseline
Combined ER+TR	Single "non-drug" trajectory	`non_drug = week * rate`

Simplified options:

Recommendation: Keep BR with biomarker moderation (core hypothesis). Consider dropping TR initially; add back if baseline drift is important.

3.2 Statistical Analysis

3.2.1 Model Choice

Current: `nlme::lme()` with CAR(1) correlation structure.

Simplified options:

Option	Formula	Pros	Cons
ANCOVA	`lm(post ~ pre + drug * biomarker)`	Fast, familiar, closed-form	Requires summary measure
t-test on change	`t.test(delta ~ group)`	Simplest	Ignores biomarker
GEE	`geepack::geeglm()`	Robust to correlation misspec.	Slightly complex
Random intercept LMM	`lme4::lmer(y ~ x + (1|id))`	Simpler than full lme	Still mixed model
Simple linear regression	`lm(response ~ on_drug * biomarker)`	Ignores clustering	Biased SE but fast
RM-ANOVA	`aov(y ~ bm_group * drug + Error(id/drug))`	Familiar to clinicians	Requires dichotomized biomarker (∼36% information loss); sphericity assumption

Recommendation: Use **ANCOVA on change scores** as primary analysis:

```
summary_data <- trial_data |>
  group_by(participant_id, on_drug) |>
  summarise(mean_response = mean(response), .groups = "drop") |>
  pivot_wider(names_from = on_drug, values_from = mean_response) |>
  mutate(change = `TRUE` - `FALSE`)

model <- lm(change ~ biomarker, data = summary_data)
p_value <- coef(summary(model))["biomarker", "Pr(>|t|)"]
```

3.2.2 Handling Repeated Measures

Current: Full longitudinal model with all timepoints.

Simplified options:

Option	Description
Summary statistics	Mean/AUC during each phase
Last observation	Use final timepoint only
First vs. last	Compare endpoints
Phase means	Average within treatment phases

Recommendation: Phase means – average response during on-drug vs. off-drug periods, then analyze the difference.

3.3 Mathematical Specification of the Interaction Test

This section formally defines the biomarker-treatment interaction test under each analysis method and connects the precision of the interaction estimate to trial design features: the number of subjects (N), the number of assessment timepoints per subject (T), the number of within-subject treatment transitions (crossovers), and the proportion of time spent on vs. off drug.

3.3.1 Notation

Symbol	Definition
Y_{it}	Observed outcome for subject i at timepoint t
bm_i	Baseline biomarker value for subject i (time-invariant)
D_{it}	Drug exposure indicator (1 = on drug, 0 = off drug)
t	Time (weeks since baseline)
u_i	Subject-specific random intercept, $u_i \sim N(0, \sigma_u^2)$
ε_{it}	Residual error, $\varepsilon_{it} \sim N(0, \sigma_\varepsilon^2)$
N	Number of subjects
T_i	Total assessment timepoints for subject i
$n_{i,\text{on}}$	Number of on-drug assessments for subject i
$n_{i,\text{off}}$	Number of off-drug assessments for subject i
K_i	Number of treatment transitions for subject i

Throughout, the **target hypothesis** is:

$$H_0 : \text{the biomarker does not predict the magnitude of the drug-specific response.}$$

The mathematical form of this hypothesis differs depending on whether the design includes within-subject drug variation.

3.3.2 The Fundamental Distinction: Within-Subject Drug Variation

Designs with within-subject drug variation (OL+BDC, CO, N-of-1): Subject i has observations both on drug ($D_{it} = 1$) and off drug ($D_{it} = 0$). The interaction is $bm_i \times D_{it}$, testing whether the biomarker predicts the within-subject drug effect.

Designs without within-subject drug variation (OL): All observations for subject i have $D_{it} = 1$. There is no within-subject drug contrast. The interaction must be reformulated as $bm_i \times t$, testing whether the biomarker predicts the rate of improvement over time while on drug. This is a fundamentally weaker test because temporal improvement confounds BR, ER, and TV.

This distinction is the primary reason the open-label design has low power (Hendrickson Figure 4A): the interaction term targets a different, noisier quantity.

3.3.3 Method 1: ANCOVA on Phase-Mean Change Scores

Step 1. Collapse repeated measures into within-subject phase means:

$$\bar{Y}_{i,\text{on}} = \frac{1}{n_{i,\text{on}}} \sum_{t:\, D_{it}=1} Y_{it}, \qquad \bar{Y}_{i,\text{off}} = \frac{1}{n_{i,\text{off}}} \sum_{t:\, D_{it}=0} Y_{it}$$

Step 2. Compute the within-subject drug effect:

$$\Delta_i = \bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}}$$

Step 3. Regress the drug effect on the biomarker:

$$\Delta_i = \gamma_0 + \gamma_1 \cdot bm_i + e_i$$

Interaction test: $H_0 : \gamma_1 = 0$ via t-test on $\hat{\gamma}_1$, with $N - 2$ degrees of freedom.

Precision of $\hat{\gamma}_1$:

$$\text{Var}(\hat{\gamma}_1) = \frac{\sigma^2_\Delta}{N \cdot \text{Var}(bm)}$$

where $\sigma^2_\Delta = \text{Var}(\Delta_i)$ is the between-subject variance of the within-subject drug effect. This variance decomposes as:

$$\sigma^2_\Delta = \text{Var}(\bar{Y}_{i,\text{on}}) + \text{Var}(\bar{Y}_{i,\text{off}}) - 2\,\text{Cov}(\bar{Y}_{i,\text{on}}, \bar{Y}_{i,\text{off}})$$

Each phase-mean variance is approximately:

$$\text{Var}(\bar{Y}_{i,\text{phase}}) \approx \sigma^2_u + \frac{\sigma^2_\varepsilon}{n_{i,\text{phase}}}$$

The covariance term arises from the shared random intercept u_i and equals σ^2_u under compound symmetry. Substituting:

$$\sigma^2_\Delta \approx \sigma^2_\varepsilon \left(\frac{1}{n_{i,\text{on}}} + \frac{1}{n_{i,\text{off}}} \right)$$

The random intercept cancels in the difference, which is a key advantage of within-subject contrasts. What remains is purely residual variance, reduced by the number of observations in each phase.

Design implications:

Design-specific behavior:

- **OL:** $n_{i,\text{off}} = 0$, so Δ_i is undefined. The ANCOVA must be reformulated as $\bar{Y}_i = \gamma_0 + \gamma_1 \cdot bm_i + e_i$ (regress total response on biomarker), which confounds BR with ER and TV.
- **OL+BDC:** $n_{i,\text{on}} \approx 5$, $n_{i,\text{off}} \approx 2$. Imbalanced phases inflate σ^2_Δ.
- **CO:** $n_{i,\text{on}} \approx 4$, $n_{i,\text{off}} \approx 4$. Balanced phases. One crossover transition.
- **N-of-1:** $n_{i,\text{on}} \approx 6$, $n_{i,\text{off}} \approx 4$. Multiple crossover transitions. Highest total observation count and reasonable balance.

Design feature	Effect on $\mathrm{Var}(\hat{\gamma}_1)$
More subjects ($N \uparrow$)	Direct reduction: $\propto 1/N$
More on-drug assessments ($n_{\text{on}} \uparrow$)	Reduces σ^2_Δ via $1/n_{\text{on}}$ term
More off-drug assessments ($n_{\text{off}} \uparrow$)	Reduces σ^2_Δ via $1/n_{\text{off}}$ term
Balanced phases ($n_{\text{on}} \approx n_{\text{off}}$)	Minimizes the harmonic mean, hence σ^2_Δ
More crossovers ($K \uparrow$)	No direct effect beyond increasing n_{on} and n_{off}; distributing observations across more transitions reduces sensitivity to period effects
Wider biomarker range ($\mathrm{Var}(bm) \uparrow$)	Direct reduction: $\propto 1/\mathrm{Var}(bm)$

OL special case. Without drug variation, the ANCOVA tests a different quantity. One approach is to use the slope of improvement over time as the subject-level summary:

$$b_i = \text{slope of } Y_{it} \text{ vs. } t \text{ for subject } i$$

$$b_i = \gamma_0 + \gamma_1 \cdot bm_i + e_i$$

Here γ_1 tests whether the biomarker predicts the rate of improvement. This conflates BR rate with ER and TV rates, explaining the poor power of the OL design.

3.3.4 Method 2: Paired t-test on Change Scores

The paired t-test applies to the same Δ_i as ANCOVA but tests a different hypothesis:

$$H_0 : E[\Delta_i] = 0 \quad \text{(no average drug effect)}$$

$$t = \frac{\bar{\Delta}}{s_\Delta/\sqrt{N}}, \quad df = N - 1$$

This does not test the biomarker interaction. It tests whether the drug works on average, collapsing over biomarker values. To recover a biomarker interaction test from a t-test framework, one must dichotomize:

$$\text{Split subjects into } bm_{\text{high}} \text{ and } bm_{\text{low}} \text{ by median}$$

$$t = \frac{\bar{\Delta}_{\text{high}} - \bar{\Delta}_{\text{low}}}{s_p\sqrt{1/N_{\text{high}} + 1/N_{\text{low}}}}, \quad df = N - 2$$

This is equivalent to ANCOVA with a binary biomarker and inherits the same design dependencies. Dichotomization discards information, reducing power relative to ANCOVA with a continuous biomarker by approximately $2/\pi \approx 64\%$ relative efficiency (Cohen 1983).

Design implications: Same as ANCOVA, but uniformly less powerful due to dichotomization. Not recommended when the biomarker is continuous.

3.3.5 Method 3: GEE (Generalized Estimating Equations)

Model (population-averaged):

$$E[Y_{it}] = \beta_0 + \beta_1 \cdot bm_i + \beta_2 \cdot D_{it} + \beta_3 \cdot t + \beta_4 \cdot bm_i \cdot D_{it}$$

with working correlation matrix $R_i(\alpha)$ and sandwich (robust) variance estimator.

Interaction test: $H_0 : \beta_4 = 0$ via Wald test using the sandwich-estimated variance:

$$z = \frac{\hat{\beta}_4}{\widehat{\mathrm{SE}}_{\mathrm{robust}}(\hat{\beta}_4)}$$

Precision of $\hat{\beta}_4$:

The sandwich estimator is consistent regardless of the working correlation structure, but efficiency depends on how well the working correlation approximates the truth. Under a correct working correlation:

$$\mathrm{Var}(\hat{\beta}_4) \approx \frac{\sigma^2_{\mathrm{eff}}}{N \cdot \mathrm{Var}(bm) \cdot \mathrm{Var}_{\mathrm{within}}(D)}$$

where $\mathrm{Var}_{\mathrm{within}}(D)$ is the within-subject variance of drug exposure and σ^2_{eff} is the effective residual variance after accounting for the working correlation.

The **within-subject variance of drug exposure** is the critical design-dependent quantity:

$$\mathrm{Var}_{\mathrm{within}}(D_i) = \bar{D}_i(1 - \bar{D}_i)$$

where $\bar{D}_i = n_{i,\mathrm{on}}/T_i$ is the proportion of time on drug. This is maximized when $\bar{D}_i = 0.5$ (equal time on and off drug) and equals zero when $\bar{D}_i = 1$ (the OL design).

Design implications:

Design feature	Effect on $\mathrm{Var}(\hat{\beta}_4)$
More subjects ($N \uparrow$)	Direct reduction: $\propto 1/N$
More timepoints ($T \uparrow$)	Reduces σ^2_{eff}; modest gain under strong autocorrelation
Balanced drug exposure ($\bar{D} \to 0.5$)	Maximizes $\mathrm{Var}_{\mathrm{within}}(D)$, directly reducing $\mathrm{Var}(\hat{\beta}_4)$
More crossovers ($K \uparrow$)	Does not change $\mathrm{Var}_{\mathrm{within}}(D)$ directly, but distributes drug variation across time
Autocorrelation ($\rho \uparrow$)	Increases effective variance; nearby observations contribute less independent information

Design-specific behavior:

- **OL:** $\mathrm{Var}_{\mathrm{within}}(D) = 0$. The interaction $bm \times D$ is not estimable. Must substitute $bm \times t$.
- **OL+BDC:** $\bar{D} \approx 0.75$. $\mathrm{Var}_{\mathrm{within}}(D) \approx 0.19$. Suboptimal but nonzero.
- **CO:** $\bar{D} = 0.5$. $\mathrm{Var}_{\mathrm{within}}(D) = 0.25$. Optimal balance.
- **N-of-1:** $\bar{D} \approx 0.55$. $\mathrm{Var}_{\mathrm{within}}(D) \approx 0.25$. Near-optimal, with multiple transitions.

GEE vs. LMM: GEE estimates population-averaged effects (marginal model), while LMM estimates subject-specific effects (conditional model). For the linear case with continuous outcomes, β_4 has the same interpretation under both approaches. The practical difference is that GEE provides

valid inference under working-correlation misspecification (via the sandwich estimator), while LMM requires correct specification of the random effects structure but is more efficient when correctly specified.

3.3.6 Method 4: Random Intercept LMM (`lme4::lmer`)

Model (subject-specific):

$$Y_{it} = \beta_0 + \beta_1 \cdot bm_i + \beta_2 \cdot D_{it} + \beta_3 \cdot t + \beta_4 \cdot bm_i \cdot D_{it} + u_i + \varepsilon_{it}$$

$$u_i \sim N(0, \sigma_u^2), \qquad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2)$$

Interaction test: $H_0 : \beta_4 = 0$ via Wald test (or Kenward-Roger / Satterthwaite F-test for small N).

Precision of $\hat{\beta}_4$:

Under the random-intercept model with independent residuals, the information matrix for β_4 depends on the effective within-subject information after projecting out the random intercept:

$$\mathrm{Var}(\hat{\beta}_4) \approx \frac{\sigma_\varepsilon^2}{\sum_{i=1}^{N} \widetilde{bm_i}^{\,2} \cdot \mathrm{SS}_{\text{within}}(D_i)}$$

where $\widetilde{bm}_i = bm_i - \overline{bm}$ is the centered biomarker value and:

$$\mathrm{SS}_{\text{within}}(D_i) = \sum_{t=1}^{T_i} (D_{it} - \bar{D}_i)^2 = T_i \cdot \bar{D}_i(1 - \bar{D}_i)$$

is the within-subject sum of squares of drug exposure for subject i. This quantity is the total "information" about drug variation contained in subject i's data.

Design implications:

Design feature	Effect on $\mathrm{Var}(\hat{\beta}_4)$	Mechanism
$N \uparrow$	$\propto 1/N$	More independent contrasts
$T \uparrow$	$\propto 1/T$ (given fixed $\bar{D}$)	Each timepoint adds to $\mathrm{SS}_{\text{within}}(D_i)$
$\bar{D} \to 0.5$	Maximizes $\bar{D}(1 - \bar{D})$	Equal on/off time maximizes contrast
$K \uparrow$	No direct effect under independence	Under AR(1), more transitions reduce autocorrelation penalty
Wider biomarker range	Increases $\sum \widetilde{bm_i}^{\,2}$	More leverage for the interaction
$\rho > 0$	Inflates effective σ_ε^2	Correlated residuals reduce effective sample size

The autocorrelation-crossover interaction. Under AR(1) residuals (as implemented in the full `nlme::lme` model with `corCAR1`), the effective information from each on-drug or off-drug block depends on block length L and autocorrelation ρ:

$$\text{effective observations per block} \approx \frac{L}{1 + (L-1)\rho} \cdot L$$

Short blocks (high K, low L) suffer less from autocorrelation than long blocks (low K, high L) because fewer consecutive observations are highly correlated. This is why the N-of-1 design (four 4-week blocks) can outperform the crossover (two 10-week blocks) when autocorrelation is moderate.

Conversely, when the drug has substantial carryover, each transition introduces a contaminated observation at the block boundary. More transitions ($K \uparrow$) means more contaminated observations. This is the mechanism behind the "precipitous decline in power" with carryover that Hendrickson et al. report (their Figure 4B).

Design-specific $\mathbf{SS_{within}}(D)$:

Design	T	n_{on}	n_{off}	$\bar{D}$	$\text{SS}_{\text{within}}(D)$	K
OL	8	8	0	1.0	0	0
OL+BDC	8	∼6	∼2	∼0.75	∼1.5	1
CO	8	4	4	0.5	2.0	1
N-of-1	12	∼7	∼5	∼0.58	∼2.9	3

The N-of-1 design has the highest $\text{SS}_{\text{within}}(D)$ due to both more total timepoints and near-balanced drug exposure. The OL design has zero within-subject drug information.

3.3.7 Method 5: Simple Linear Regression (Ignoring Clustering)

Model:

$$Y_{it} = \beta_0 + \beta_1 \cdot bm_i + \beta_2 \cdot D_{it} + \beta_3 \cdot t + \beta_4 \cdot bm_i \cdot D_{it} + \varepsilon_{it}$$

This is the same fixed-effects structure as Method 4 but without the random intercept u_i. All $N \times T$ observations are treated as independent.

Interaction test: $H_0 : \beta_4 = 0$ via OLS t-test.

Precision of $\hat{\beta}_4$ (nominal):

$$\text{Var}_{\text{OLS}}(\hat{\beta}_4) = \frac{\hat{\sigma}^2}{\text{SS}(bm \cdot D \mid \text{other predictors})}$$

where $\hat{\sigma}^2$ is the OLS residual variance. Because the model ignores the random intercept, $\hat{\sigma}^2$ absorbs both σ_u^2 and σ_ε^2, but the denominator treats all $N \times T$ observations as independent. This produces **downward-biased standard errors** because the effective sample size for the between-subject component (bm_i) is N, not $N \times T$.

Type I error inflation: The degree of inflation depends on $\sigma_u^2/\sigma_\varepsilon^2$ (the intraclass correlation) and T. With $T = 8$ and ICC $= 0.3$, the nominal $\alpha = 0.05$ test has an actual Type I error of approximately 0.15–0.25.

Design implications: Same as Method 4 for power ordering across designs, but with inflated Type I error that invalidates nominal p-values. Not recommended for inference but can serve as a fast screening tool.

3.3.8 Method 6: Repeated Measures ANOVA (Biomarker Dichotomized)

Setup. Because repeated measures ANOVA requires a grouping factor (not a continuous covariate), the biomarker must be dichotomized:

$$G_i = \begin{cases} 1 & \text{if } bm_i > \text{median}(bm) \\ 0 & \text{otherwise} \end{cases}$$

The data are arranged as a $N \times T$ matrix of outcomes, with two between-subject groups ($G = 0, 1$) and a within-subject factor D (on-drug vs. off-drug). If the design has multiple timepoints within each drug condition, there is an additional within-subject factor for time nested within drug phase.

Model (split-plot / mixed ANOVA):

$$Y_{it} = \mu + \alpha_{G_i} + \delta_{D_{it}} + (\alpha\delta)_{G_i,D_{it}} + \pi_i + \varepsilon_{it}$$

where μ is the grand mean, α_{G_i} is the between-subject biomarker group effect, $\delta_{D_{it}}$ is the within-subject drug effect, $(\alpha\delta)_{G_i,D_{it}}$ is the biomarker-by-drug interaction, $\pi_i \sim N(0, \sigma_\pi^2)$ is the subject effect (random), and $\varepsilon_{it} \sim N(0, \sigma_\varepsilon^2)$ is the residual.

Interaction test: $H_0 : (\alpha\delta)_{G,D} = 0$ for all combinations, tested via:

$$F = \frac{\text{MS}_{G\times D}}{\text{MS}_{\text{error(within)}}}, \quad df_1 = (g-1)(d-1), \quad df_2 = (N-g)(d-1)$$

where $g = 2$ (biomarker groups) and $d = 2$ (on-drug, off-drug), giving $df_1 = 1$ and $df_2 = N - 2$.

Sphericity. When more than two levels of the within-subject factor are present (e.g., multiple timepoints within each drug phase), the F-test assumes sphericity: equal variances of all pairwise differences between within-subject levels. Violation inflates the Type I error rate. The Greenhouse-Geisser (GG) and Huynh-Feldt (HF) corrections multiply both df_1 and df_2 by a correction factor $\hat{\varepsilon} \in [1/(k-1), 1]$:

$$df_1' = df_1 \cdot \hat{\varepsilon}, \qquad df_2' = df_2 \cdot \hat{\varepsilon}$$

When the within-subject factor is simply on-drug vs. off-drug ($k = 2$), sphericity is satisfied trivially (there is only one pairwise difference), and no correction is needed.

Precision of the interaction test. For the simple case ($g = 2$ groups, $d = 2$ drug conditions, collapsing timepoints within phases into means):

$$\text{MS}_{G\times D} = \frac{N}{4}\left(\bar{\Delta}_{\text{high}} - \bar{\Delta}_{\text{low}}\right)^2$$

$$\text{MS}_{\text{error(within)}} = \frac{1}{N-2}\sum_{i=1}^{N}(\Delta_i - \bar{\Delta}_{G_i})^2$$

where $\Delta_i = \bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}}$ as in the ANCOVA approach. The noncentrality parameter for the F-test is:

$$\lambda = \frac{N}{4} \cdot \frac{(\mu_{\Delta,\text{high}} - \mu_{\Delta,\text{low}})^2}{\sigma_\Delta^2}$$

Comparison to ANCOVA (Method 1). The repeated measures ANOVA on phase means with dichotomized biomarker is algebraically equivalent to a two-sample t-test on Δ_i between biomarker

groups. The ANCOVA on change scores with continuous biomarker tests the same conceptual hypothesis but retains the full biomarker distribution. The efficiency loss from dichotomization is well established: for a normally distributed biomarker, the median split retains only $2/\pi \approx 63.7\%$ of the information, corresponding to a relative efficiency of approximately 0.64 (Cohen 1983, Royston et al. 2006). In sample-size terms, the repeated measures ANOVA requires approximately $N/0.64 \approx 1.57N$ subjects to achieve the same power as ANCOVA with a continuous biomarker.

When timepoints are not collapsed. If the full set of T timepoints is retained as within-subject levels, the model becomes a two-way mixed ANOVA with G (between) and timepoint (within). The interaction $G \times$ timepoint tests whether the two biomarker groups have different temporal profiles. This formulation:

- Gains power from using all observations (no information loss from averaging)
- Loses specificity by testing a composite hypothesis across all timepoints rather than the drug contrast specifically
- Requires the sphericity assumption across all T timepoints, which is typically violated in longitudinal data and necessitates GG/HF correction
- The corrected degrees of freedom can substantially reduce power when $\hat{\varepsilon}$ is small (strong autocorrelation, many timepoints)

Design implications:

Design feature	Effect on power	Mechanism
$N \uparrow$	$\propto N$ (via λ)	More independent group comparisons
More on/off-drug assessments	Reduces σ^2_Δ when collapsing to phase means	More precise Δ_i
Balanced phases	Same benefit as ANCOVA	Minimizes $\mathrm{Var}(\Delta_i)$
Balanced biomarker groups	Maximizes F-test power	Median split guarantees this
$K \uparrow$	Indirect benefit via $n_{\text{on}}, n_{\text{off}}$	Same as ANCOVA
Timepoints retained as levels	Increases df_1 but triggers sphericity issues	GG/HF correction can offset gains
$\rho \uparrow$	Reduces $\hat{\varepsilon}$ when timepoints retained	Corrected df shrink

Design-specific behavior:

- **OL:** No within-subject drug factor. The within-subject factor becomes time, and the interaction $G \times$ time tests whether biomarker groups have different slopes. Same confounding problem as all other methods.
- **OL+BDC:** The within-subject drug factor has two levels but is heavily imbalanced. The sparse off-drug phase inflates σ^2_Δ.
- **CO:** Clean two-level within-subject drug factor with balanced phases. Sphericity trivially satisfied. The RM-ANOVA is most natural here.
- **N-of-1:** Multiple drug-on and drug-off blocks. If collapsed to phase means, behaves like ANCOVA with good balance. If blocks are retained as separate levels, sphericity becomes a concern.

Summary. Repeated measures ANOVA is conceptually accessible and widely implemented, making it attractive for clinical audiences. However, for the biomarker interaction test specifically, it suffers

from two compounding limitations: (1) dichotomization of the biomarker discards approximately 36% of the information, and (2) when timepoints are retained rather than collapsed, sphericity violations and GG/HF corrections can further erode power. For these reasons, ANCOVA on change scores (Method 1) or the random intercept LMM (Method 4) are generally preferable when the biomarker is continuous.

3.3.9 The OL Design: A Fundamentally Different Interaction Test

For the open-label design, all six methods face the same problem: there is no within-subject drug variation. The interaction must be formulated as biomarker-by-time:

$$Y_{it} = \beta_0 + \beta_1 \cdot bm_i + \beta_2 \cdot t + \beta_3 \cdot bm_i \cdot t + u_i + \varepsilon_{it}$$

Here β_3 tests whether the biomarker predicts the slope of improvement over time. This is a weaker test than β_4 (the biomarker-by-drug interaction) for two reasons:

1. **Confounding.** The slope dY/dt includes BR, ER, and TV. The biomarker-by-time interaction conflates all three.
2. **Precision.** The within-subject information for the time slope depends on $\mathrm{SS}_{\text{within}}(t) = \sum(t-\bar{t})^2$, which is typically smaller than $\mathrm{SS}_{\text{within}}(D)$ for designs with balanced crossovers.

3.3.10 Summary: Interaction Test Precision Across Designs and Methods

Table 1 summarizes the effective information for the biomarker interaction test across all design-method combinations. Entries are proportional to $1/\operatorname{Var}(\hat{\beta}_{\text{interaction}})$ (higher = more power).

Table 1: Interaction test properties by design and method.

	OL	OL+BDC	CO	N-of-1
Interaction term	$bm \times t$	$bm \times D$	$bm \times D$	$bm \times D$
Target	Total improvement rate	Drug-specific effect	Drug-specific effect	Drug-specific effect
Confounded with	ER, TV slopes	–	–	–
ANCOVA info	$N \cdot \mathrm{Var}(bm) \cdot \mathrm{Var}(b_i)^{-1}$	$N \cdot \mathrm{Var}(bm) \cdot \sigma_\Delta^{-2}$	$N \cdot \mathrm{Var}(bm) \cdot \sigma_\Delta^{-2}$	$N \cdot \mathrm{Var}(bm) \cdot \sigma_\Delta^{-2}$
LMM info	$\sum \widetilde{bm}_i^2 \cdot \mathrm{SS}(t_i)$	$\sum \widetilde{bm}_i^2 \cdot 1.5$	$\sum \widetilde{bm}_i^2 \cdot 2.0$	$\sum \widetilde{bm}_i^2 \cdot 2.9$
GEE info	$\approx$ LMM	$\approx$ LMM	$\approx$ LMM	$\approx$ LMM
RM-ANOVA info	$\approx 0.64\times$ ANCOVA	$\approx 0.64\times$ ANCOVA	$\approx 0.64\times$ ANCOVA	$\approx 0.64\times$ ANCOVA
K **(transitions)**	0	1	1	3
Autocorr. vuln.	Low	Medium	Medium	Low per block, more boundary effects
Carryover vuln.	None	Low (1 transition)	Low (1 transition)	High (3 transitions)

The RM-ANOVA row reflects the $\approx 2/\pi$ efficiency penalty from dichotomizing the biomarker at the median. If timepoints within phases are retained as separate within-subject levels rather than collapsed to phase means, GG/HF corrections further reduce effective degrees of freedom under autocorrelation.

The N-of-1 design achieves the highest information under the LMM and GEE when carryover is absent, owing to both its larger total T and near-balanced drug exposure. The crossover achieves optimal balance ($\bar{D} = 0.5$) with fewer transitions, making it more robust to carryover at the cost of fewer total timepoints.

3.4 Trial Designs (Non-Simplifiable)

Critical: The four Hendrickson trial designs must be preserved exactly as specified. The design structure is the core scientific comparison and cannot be simplified.

Design	Weeks	Structure	Blinding
Open-Label (OL)	1–8	All on drug, unblinded	None
OL + Blinded Disc. (OL+BDC)	1–4 OL, 5–8 randomized	Partial	
Traditional Crossover (CO)	4 wks drug, 4 wks placebo (order rand.)	Full	
N-of-1 (Hybrid)	1–4 OL, 5–12 two 4-wk crossover cycles	Partial	

Each design must specify: (1) week-by-week `on_drug` indicator, (2) blinding status (affects expectancy response), and (3) randomization path (which weeks are randomized).

```
designs <- list(
  ol = list(weeks = 1:8,
    on_drug = rep(TRUE, 8), blinded = rep(FALSE, 8),
    expectancy = rep(TRUE, 8)),
  ol_bdc = list(weeks = 1:8,
    on_drug = c(rep(TRUE, 4), NA, NA, NA, NA),
    blinded = c(rep(FALSE, 4), rep(TRUE, 4)),
    expectancy = c(rep(TRUE, 4), rep(FALSE, 4))),
  crossover = list(weeks = 1:8,
    on_drug = NA, blinded = rep(TRUE, 8),
    expectancy = rep(FALSE, 8)),
  nof1 = list(weeks = 1:12,
    on_drug = c(rep(TRUE, 4), rep(NA, 8)),
    blinded = c(rep(FALSE, 4), rep(TRUE, 8)),
    expectancy = c(rep(TRUE, 4), rep(FALSE, 8))))
```

`NA` values indicate randomized assignment at simulation time.

3.5 Carryover Effects

Current: Exponential decay with half-life parameter.

Simplified options:

Option	Formula	Description
No carryover	carryover $= 0$	Assume instant washout
Binary carryover	$\mathbf{1}[t_{\text{since}} < \text{washout}]$	All-or-nothing
Linear decay	$\max(0, 1 - t_{\text{since}}/\text{washout})$	Linear decrease
Step function	$0.5 \cdot \mathbf{1}[t_{\text{since}} < 1]$	Partial for one period

Recommendation: Start with **no carryover**, add **linear decay** as sensitivity analysis.

3.6 Censoring/Dropout

Current: Five patterns (none, balanced, high dropout, more biased, more flat) with complex probability functions.

Simplified options:

Option	Description
No dropout	Complete data for all
MCAR	Random dropout, constant probability
Single MAR	One dropout pattern tied to response

Recommendation: Start with **no dropout**. Add MCAR (e.g., 10% per timepoint) as sensitivity.

3.7 Parameter Space

Parameter	Current	Simplified
BR_rate	0.05, 0.15, 0.3	0.1, 0.3 (2 levels)
ER_rate	0.05, 0.15, 0.3	0.1 (fixed)
TR_rate	0.05, 0.15, 0.3	0 (removed)
biomarker_mod	0, 0.3, 0.6	0.3 (fixed), vary for sensitivity
carryover	0, 0.1, 0.2	0 (default), 0.2 (sensitivity)
censoring	5 levels	none (default)
N	35, 70	35, 50, 70

Recommendation: Single main simulation with key parameters, separate one-at-a-time sensitivity analyses.

4 Implementation Strategy

Phase 1: Minimal Viable Simulation. Single script `simulation.R` (∼200 lines); random intercept model for data generation; ANCOVA on phase means for analysis; four designs with simplified specification; no carryover, no dropout; fixed BR_rate, ER_rate; vary biomarker_mod and N.

Phase 2: Add Complexity as Needed. Add carryover (linear decay); add dropout (MCAR); sensitivity on BR_rate; compare ANCOVA vs. LMM results.

Phase 3: Validation. Compare power estimates to pmsimstats2025; document where simplifications matter vs. do not.

5 Code Architecture

```
01b-pmsimstats-simple/
  R/
    functions.R            # Shared utility functions
  analysis/
    scripts/
      simulation.R         # Main simulation (single file)
```

```
    sensitivity.R        # Parameter sensitivity (optional)
  output/
docs/
  simplification-plan.md
  simplification-plan.tex
```

Key design principles: (1) Single file first. (2) Explicit over clever: write out loops rather than complex abstractions. (3) Base R preferred; minimize dependencies. (4) No parallelization by default. (5) Inline parameters at top of script.

6 Expected Outcomes

Metric	pmsimstats2025	pmsimstats-simple (target)
Lines of code	∼700 per script	∼200 total
Dependencies	nlme, lme4, MASS, zoo, patchwork, future, furrr	tidyverse only
Runtime (20 iter)	∼160 sec	∼10–20 sec
Convergence issues	Occasional lme failures	None (closed-form)
Parameters	∼15 varying	∼5 varying

7 Literature Support for Simplifications

7.1 Key Reference: Hendrickson et al. (2020)

The primary reference is Hendrickson et al. (2020), "Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation," *Front. Digit. Health* 2:13.

Key methodological points:

1. **Three-factor response model:** BR, ER, TR, each modeled via Gompertz function with max, rate, displacement parameters.
2. **Analysis approach:** LME (`lme4`) with random intercept per subject. For designs with on/off drug: `response ~ on_drug + biomarker + on_drug:biomarker + (1|id)`. For open-label only: `response ~ time + biomarker + time:biomarker + (1|id)`.
3. **Carryover:** Exponential decay with half-life parameter. Critical finding: even short (0.1 week) carryover causes "precipitous decline in power" for N-of-1 and OL+BDC designs.
4. **Censoring/dropout:** Probability $= \beta_0 + \beta_1 \times (\text{change in symptoms})^2$. Biased dropout where responders more likely to drop out during placebo.
5. **Key result:** N-of-1 design has power between OL+BDC (lower) and traditional crossover (higher), but allows all participants to start on open-label treatment.

7.2 Analysis Method Comparisons

Chen & Chen (2014) compared paired t-test, mixed effects model, mixed effects model of difference, and meta-analysis for N-of-1 trials (3–4 cycles, $N = 1$–30):

Recommendation: "Paired t-test was recommended to use for normally distributed data of N-of-1 trials irrespective with or without carryover effect."

Method	Type I Error	Power	Bias	MSE
Paired t-test	Near nominal	Highest	Comparable	Smallest
Mixed effects	Similar	Lower	Smaller	Bigger
ME of difference	Far from nominal	Low	Large	Large
Meta-analysis	Far from nominal	Low	Large	Large

Critique (Araujo et al. 2016): Chen & Chen models "do not include a treatment by patient interaction" advocated in meta-analysis literature.

7.3 ANCOVA for RCTs

BMC Medical Research Methodology (2014): "ANCOVA should be the analysis of choice, a priori, for RCTs with a single post-treatment outcome measure previously measured at baseline; its superiority is particularly marked when baseline imbalance is present."

7.4 Biomarker-Treatment Interaction Power

Trials (2018): "Tests of interactions are often lacking statistical power." "The probability of detecting a biomarker-treatment interaction can be increased by including prognostic variables."

7.5 N-of-1 Power Analysis

Wang & Schork (2019): Analytical and simulation-based power analysis for N-of-1 designs considering number of cycles, serial correlation, washout periods, heteroscedasticity, and carryover.

7.6 Implications for Simplification

Literature Finding	Implication for pmsimstats-simple
Paired t-test has highest power (Chen 2014)	ANCOVA on change scores is reasonable
ANCOVA superior for baseline-adjusted outcomes (BMC 2014)	Supports phase-mean approach
Carryover devastating to N-of-1 power (Hendrickson 2020)	Must include carryover sensitivity
LME may have lower power than simpler methods	Justifies avoiding complex mixed models
Interaction tests lack power (Trials 2018)	Need adequate sample sizes

8 Decision Log

9 References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA (2020). Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation. *Front. Digit. Health* 2:13. https://doi.org/10.3389/fdgth.2020.00013
2. Chen X, Chen P (2014). A Comparison of Four Methods for the Analysis of N-of-1 Trials. *PLoS ONE* 9(2): e87752. https://doi.org/10.1371/journal.pone.0087752

Decision	Rationale	Date
Use ANCOVA	Chen 2014 shows paired t-test $\approx$ ANCOVA has highest power; BMC 2014 supports ANCOVA	2026-02-14
Keep 4 designs	Core scientific comparison per Hendrickson 2020	2026-02-14
Keep ER component	Hendrickson shows expectancy affects power	2026-02-14
Include carryover sensitivity	Hendrickson shows carryover "precipitous" effect	2026-02-14
No carryover default	Simplifies initial analysis; add as sensitivity	2026-02-14
Random intercept + iid	Chen 2014 shows simpler models perform well	2026-02-14

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